Bank Customer Churn: Evaluation of Experimental Design, PCA and Bayesian Methods

IT3081 Statistical Modelling | Group consultancy report | Member 3 — Tasks 6, 7 and 8

Prepared 18 October 2026 • Working draft: integrate with final group report and verify preprocessing before submission.

Executive summary

This section evaluates three methodological extensions for the bank customer churn consultancy problem. Task 6 proposes randomized retention experiments, Task 7 assesses principal component analysis (PCA) using the group's exploratory numerical results, and Task 8 discusses probabilistic classification, uncertainty-aware regression and decision-theoretic targeting. The dataset contains 28,382 customers, of whom 5,260 (18.53%) churned and 23,122 (81.47%) did not. These approaches address different questions: causal effectiveness of interventions, redundancy in numerical predictors, and decisions under uncertainty.

Evidence status: Task 7 includes actual exploratory results provided by the group. Task 6 is a proposed future design, not a completed experiment. Task 8 is a critical evaluation only; no Naïve Bayes or Bayesian regression estimates are claimed. Final comparisons with Member 2/4 models remain pending.

Task 6 — Critical Evaluation of Experimental Design

6.1 Purpose and rationale

Historical churn data can reveal associations between customer characteristics and account closure but generally cannot determine whether a retention intervention causes a reduction in churn. A randomized field experiment could evaluate whether a personalized contact, account-benefit offer or financial guidance genuinely changes retention. The experimental unit would be an eligible bank customer; the treatment would be a specified intervention; the primary outcome would be churn within a predefined follow-up period (for example, 90 days). Follow-up time, eligibility criteria, outcome definition and consent/privacy safeguards must be fixed before implementation.

6.2 Completely Randomized Design (CRD)

In a CRD, eligible customers are assigned at random to treatment and control groups, without first subdividing them into blocks. For example, a bank could randomly allocate 1,000 eligible customers between (A) a personalized retention offer and (B) ordinary banking services, with 500 assigned to each arm. Numbers are hypothetical and not derived from the dataset. Random allocation reduces systematic selection bias and allows differences in outcomes to be attributed to the treatment under appropriate assumptions. A suitable primary effect estimate is the difference in observed churn proportions between the two groups, accompanied by a confidence interval and a prespecified statistical test or regression estimate.

Advantages include simplicity, operational convenience, and strong causal identification if randomization, treatment delivery and follow-up are successful. Limitations include random imbalances in smaller customer segments, dilution if an offer is ineffective for many customers, noncompliance, attrition, and the possibility that customer interactions contaminate the control condition. The sample size should be determined by a prospective power calculation rather than the illustrative 1,000-person example.

6.3 Randomized Complete Block Design (RCBD)

An RCBD first groups experimental units into meaningful blocks, then randomizes every treatment within each block. For the bank, blocks might be based on occupation, baseline churn-risk band or account-balance band. For example, salaried, self-employed, student and retired customers could each receive both the offer and control condition through separate within-block random assignments. Blocking is most useful when the blocking characteristic strongly predicts the outcome; it can reduce residual variation and improve precision relative to unblocked randomization.

For a strict balanced RCBD, each block includes the full treatment set with appropriate replication. Real banking segments have unequal sizes, so a practical stratified randomized trial may be a more accurate term unless a complete block structure is enforced. Sparse strata, customer movement across segments, operational complexity and unequal take-up may limit feasibility. Effects should be estimated with block indicators and, if justified, treatment-by-block interactions to investigate heterogeneity.

6.4 Critical comparison and feasibility

Criterion	CRD	RCBD / stratified trial
Assignment	Randomize among all eligible customers	Randomize within predefined customer blocks
Strength	Simpler setup and general treatment estimate	Can improve precision if block predicts churn
Weakness	May be imbalanced within small segments	Requires valid strata, adequate sizes and extra controls
Best use	Initial bank-wide pilot	Segment-focused campaign across key risk groups

Implementation would require documented eligibility, secure randomization, CRM integration, standardized treatment delivery, a retention outcome window and protection against exposing sensitive financial information. The control arm should retain normal services; treatments should not be misleading or unfairly discriminatory. Business evaluation should track intervention cost, customer complaints and unintended outcomes as well as churn. The randomized plan-recommendation study by Ascarza, Iyengar and Schleicher (2016) demonstrates why seemingly helpful retention interventions should be tested rather than presumed effective: the treatment group in that study had higher churn than the control group. Its service setting differs from banking and therefore does not establish an expected bank effect.

6.5 Recommendation

A small CRD pilot is the more straightforward initial option if the goal is to establish whether an offer has an overall causal effect. A stratified or appropriately structured RCBD extension is recommended when preliminary evidence identifies an important segmentation variable and sufficient sample sizes are available in every block. This is a future experimental proposal: no treatment effect has been measured in the group's present project.

Task 7 — Critical Evaluation of Principal Component Analysis

7.1 Analytical objective and methodology

PCA transforms correlated numerical variables into orthogonal linear combinations ranked by explained variance. It is an unsupervised dimensionality-reduction method, not a churn classifier. The submitted exploratory run uses 13 numerical predictor columns rather than the entire original dataset. It excludes the churn target from PCA, which is methodologically appropriate. Dataset/source statistics below are taken from the accompanying CSV

outputs and figures supplied by the group. It remains necessary to confirm that missing-value handling, scaling and PCA fitting used training data only.

7.2 Evidence of redundancy and multicollinearity

Variable pair	Pearson r	Spearman rho
average_monthly_balance_prevQ / previous_month_balance	0.994	0.912
current_balance / current_month_balance	0.983	0.891
average_monthly_balance_prevQ / current_month_balance	0.976	0.858
previous_month_end_balance / current_month_balance	0.975	0.841
previous_month_end_balance / average_monthly_balance_prevQ	0.971	0.830

The near-perfect positive Pearson correlations among balance measures indicate substantial overlapping linear information. For example, average_monthly_balance_prevQ and previous_month_balance correlate at approximately 0.994. By contrast, current_month_credit and current_month_debit have Pearson r ≈ 0.94 but only moderate Spearman association (approximately 0.47), warranting caution about possible influences from outliers, ties or nonlinearity.

Feature	VIF
average_monthly_balance_prevQ	209.85
previous_month_balance	141.36
current_month_balance	93.37
current_balance	38.09
previous_month_end_balance	32.08
current_month_debit	10.33
current_month_credit	10.05

Seven of the 13 numerical predictors have VIF values greater than 10, with a maximum of 209.85. These figures strongly support further examination of redundant financial predictors. They do not, by themselves, establish that the classifier is inaccurate or that PCA will improve discrimination.

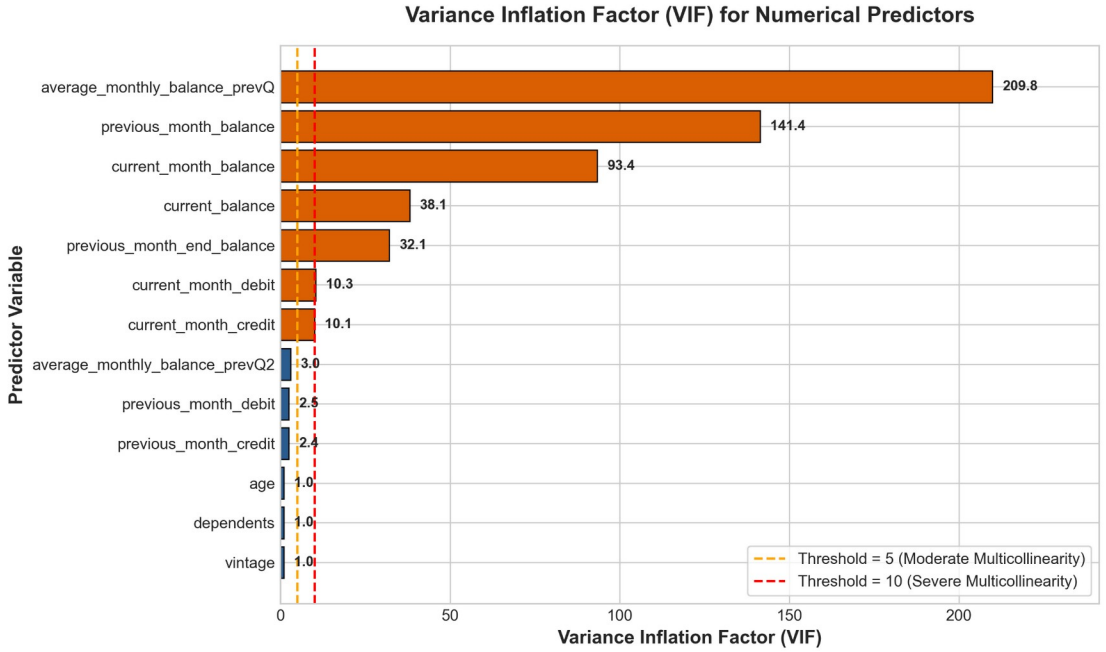


Figure 7.1. VIF results for the 13 numerical predictors. High balance-related values motivate PCA investigation.

7.3 Skewness, outliers and preprocessing

Financial feature	Skewness	Maximum
current_month_credit	143.32	12,269,845
current_month_debit	115.62	7,637,857
current_balance	99.03	5,905,904
current_month_balance	97.31	5,778,185

The financial measures display severe positive skewness and isolated extreme values. This matters because conventional PCA relies on variance and can be disproportionately influenced by very large observations, even after StandardScaler is applied. A defensible sensitivity analysis should compare appropriately transformed or robustly processed numerical features with the original PCA. Transformations must respect legitimate negative balances (for example, asinh may be preferable to log1p where negative values occur).

7.4 Principal components and retained variance

PC	Individual variance (%)	Cumulative variance (%)
PC1	42.66	42.66
PC2	16.67	59.33
PC3	11.52	70.85
PC4	7.73	78.58
PC5	7.67	86.26
PC6	7.61	93.87
PC7	3.06	96.92
PC8	1.99	98.92

The first component captures 42.66% of standardized variance. Five components preserve approximately 86.3%, six preserve 93.9%, and seven preserve 96.9%. Retaining six PCs would compress 13 numerical predictors to six constructed dimensions (a 53.8% reduction in dimension count) while losing about 6.1% of standardized numerical variance. This is not a prediction-accuracy statistic.

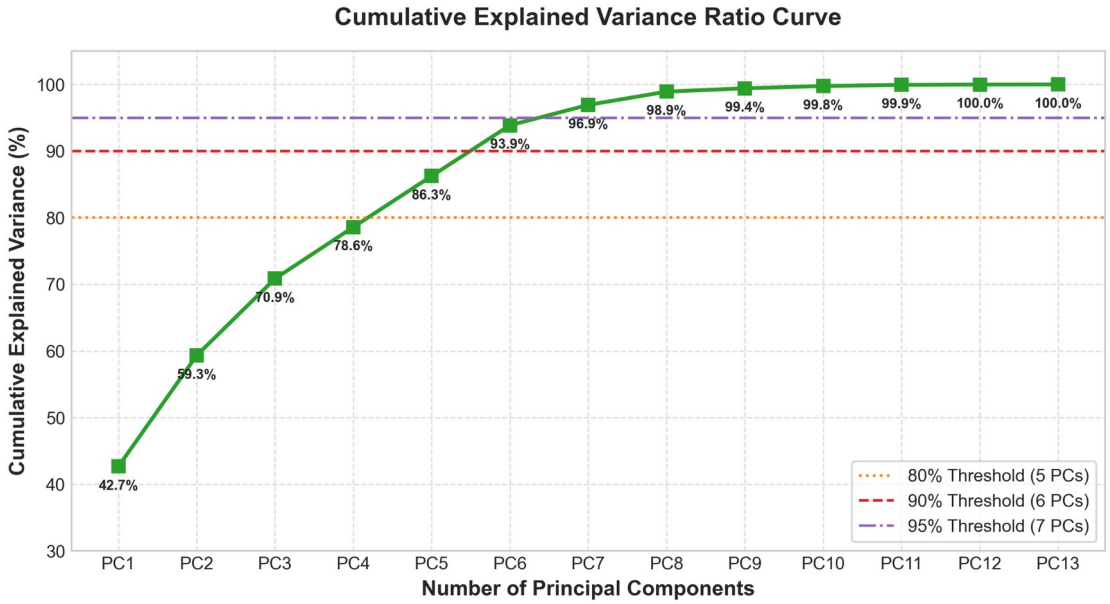


Figure 7.2. Cumulative explained variance showing thresholds and the candidate six-component solution.

7.5 Interpretation of component loadings

Feature	PC1	PC2	PC3
vintage	-0.002	-0.008	0.003
age	0.032	0.038	0.014
current_balance	0.412	-0.053	-0.034
previous_month_end_balance	0.417	-0.034	-0.013
average_monthly_balance_prevQ	0.421	-0.040	-0.017
current_month_credit	0.029	0.572	-0.417
previous_month_credit	0.057	0.406	0.575
current_month_debit	0.040	0.578	-0.401
previous_month_debit	0.070	0.402	0.574

PC1 has similar positive contributions from account-balance measures and can provisionally be interpreted as an overall liquidity/balance factor. PC2 loads positively on credit and debit flows and resembles an activity-volume dimension. PC3 contrasts current-month transaction values (negative coefficients) with prior-month transaction values (positive coefficients). These are descriptive interpretations of the PCA directions, not verified causal or churn-predictive factors. Loading signs are arbitrary up to a whole-component sign reversal.

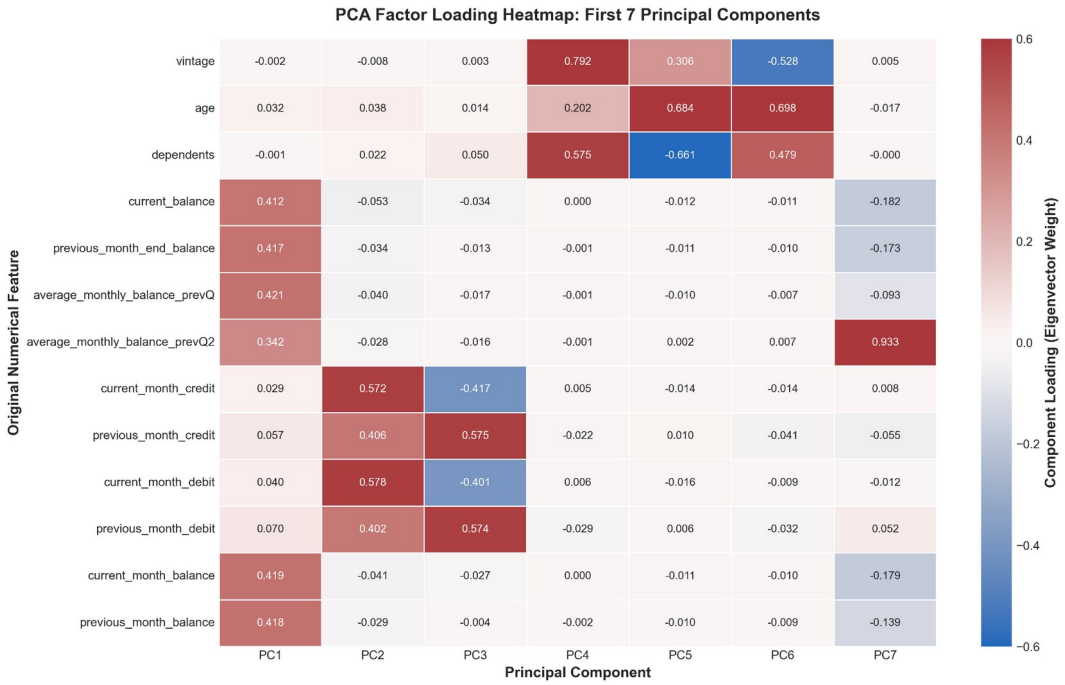


Figure 7.3. Loading coefficients for the first seven principal components.

7.6 Effect on predictive modelling and limitations

The PC1–PC2 scatterplot shows substantial visual overlap between churned and retained customers. This indicates no obvious two-dimensional separation in those particular unsupervised directions; it does not establish that PCA is unhelpful for churn prediction, because churn information may reside in later PCs or in nonlinear interactions. Dimensionality reduction can reduce redundant predictors and numerical instability, but may remove low-variance signals that matter for churn, complicate explanation to bank managers, and respond sensitively to extreme financial values.

A valid future comparison would fit preprocessing and PCA on training folds only, use the same untouched test partition as non-PCA alternatives, and assess ROC-AUC, PR-AUC, recall, precision, calibration and operational usefulness. The group's updated predictive results are not yet available, so no PCA performance improvement is claimed.

7.7 PCA recommendation

PCA is a credible candidate for this dataset, especially for the highly correlated balance variables, but is not automatically necessary given the modest overall predictor count. A targeted PCA on correlated monetary variables, with age, account tenure and other interpretable independent features kept outside the transformation, should be considered alongside full numerical PCA and interpretable regularized regression. Choose a component count based on variance retention, stability, interpretability and later out-of-sample validation; six PCs are a reasonable exploratory candidate rather than a final deployment recommendation (Jolliffe and Cadima, 2016).

Task 8 — Critical Evaluation of Bayesian Statistical Methods

8.1 Bayesian rationale

Bayesian statistics updates prior beliefs using observed data, resulting in posterior distributions. In churn analysis, a probability-based approach can support individual risk estimates, uncertainty quantification and retention decisions. Bayes' theorem is $P(H \mid D) = P(D \mid H)P(H)/P(D)$, where H represents a hypothesis and D observed evidence.

The observed churn prevalence (18.53%) provides a descriptive base rate for this sample, not a universally appropriate prior for every segment or future period.

8.2 Naïve Bayes classification

Naïve Bayes estimates class probabilities from a class prior and feature likelihoods under the simplifying assumption that predictors are conditionally independent given churn status: P(C|x) is proportional to P(C) times the product of P(x_j|C). It is simple, fast and often effective as a baseline with mixed feature likelihood formulations (for example, categorical and appropriately modelled continuous inputs). Yet the assumption is implausible for this dataset's strongly correlated balance variables, where multiple columns may repeatedly contribute the same evidence. Untested probability calibration and inappropriate Gaussian assumptions under extreme financial skew also present risks. Consequently, Naïve Bayes is suitable for evaluation but cannot currently be recommended over the group's other methods.

Boonmunewai, Chantaraksa and Rodjanadid (2024) studied bank churn with Naïve Bayes and alternative classifiers while examining sampling approaches for imbalanced data. The study supports Naïve Bayes as a legitimate comparison candidate but its measured results must not be transplanted to this dataset.

8.3 Bayesian logistic regression

Bayesian logistic regression models a binary churn outcome using a logistic link, with probability distributions placed on regression parameters. Rather than only reporting a point estimate and frequentist uncertainty interval, it produces posterior distributions and credible intervals conditional on the data, model and prior assumptions. Regularizing/weakly informative priors can reduce unstable coefficient estimates in correlated or limited-data settings; however, near-identical predictors may still impede separate interpretation, and priors do not substitute for checking the feature representation. Posterior predictive checks, convergence diagnostics and prior sensitivity analysis are essential for trustworthy use. Gelman et al. (2008) discuss weakly informative priors for logistic regression.

This is an applicability assessment only; no Bayesian logistic regression was fitted as part of the supplied Task 7 outputs. There are no posterior estimates or credible intervals to report at present.

8.4 Bayesian decision-making and retention targeting

Risk prediction and treatment choice are different tasks. A churn probability indicates a customer's likelihood of leaving, not the causal probability that an offer will prevent departure. A Bayesian decision framework would compare posterior expected utility across actions such as offer, contact without incentive or no intervention, accounting for intervention cost, potential retained-customer value, uncertain treatment efficacy and undesirable side effects. The appropriate decision is the action with the highest posterior expected utility subject to budget and fairness constraints.

For illustration only: if a retention offer costs LKR 1,000, successful retention provides LKR 10,000 net benefit before the offer cost, and the offer prevents churn with conditional probability 0.40 among customers who otherwise would leave, then expected incremental value under restrictive simplifying assumptions is p(churn) × 0.40 × 10,000 − 1,000. At p(churn)=0.30 the value is LKR 200. These invented financial inputs are not from the bank and should not be presented as empirical findings. In operational use, effectiveness must be estimated using an experimental treatment effect or other appropriately identified intervention study, rather than equating churn risk with uplift.

8.5 Critical comparison and recommendation

Method	Potential contribution	Main limitation
Naïve Bayes	Fast probabilistic churn baseline	Feature conditional-independence assumption; calibration

Bayesian logistic regression	Posterior uncertainty, regularizing priors	Prior sensitivity, computation and diagnostics
Bayesian decision theory	Cost- and benefit-aware retention actions	Requires valid costs, intervention effects and utilities

Bayesian methods are appropriate as candidate extensions because the business faces uncertain predictions and asymmetric costs. Naïve Bayes can be compared empirically after suitable preprocessing; Bayesian logistic regression can quantify parameter and predictive uncertainty; and decision theory offers a particularly strong bridge between churn probabilities and business actions. Nevertheless, no Bayesian method can be declared better without external validation and reliable economic inputs. Profit-aware churn studies also emphasize that statistically strong discrimination and business utility need not select the same targeting strategy (Verbeke et al., 2012).

Integration, evidence limitations and next steps

Task 6 remains a future experimental proposal; the current dataset is observational. Task 7 findings were calculated from uploaded exploratory CSV results, with key methodological details requiring confirmation from the notebook. Task 8 has not been implemented, so all predictive-method comparisons are pending. The group should document feature provenance and observation dates, ensure train-only preprocessing, assess skew/outlier sensitivity, and use the revised model's held-out predictions for later comparative and financial evaluation. These qualifications should be retained when merging this section into the final consultancy report.

References

Ascarza, E., Iyengar, R., & Schleicher, M. (2016). The perils of proactive churn prevention using plan recommendations: Evidence from a field experiment. Journal of Marketing Research, 53(1), 46–60. https://doi.org/10.1509/jmr.13.0483

Boonmunewai, J., Chantaraksa, T., & Rodjanadid, B. (2024). Evaluating machine learning methods for solving class imbalance in banking customer data: A comparative study. KKU Science Journal, 52(3), 349–362. https://doi.org/10.14456/kkuscij.2024.27

Gelman, A., Jakulin, A., Pittau, M. G., & Su, Y.-S. (2008). A weakly informative default prior distribution for logistic and other regression models. Annals of Applied Statistics, 2(4), 1360–1383. https://doi.org/10.1214/08-AOAS191

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Verbeke, W., Dejaeger, K., Martens, D., Hur, J., & Baesens, B. (2012). New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. European Journal of Operational Research, 218(1), 211–229.

Group exploratory analysis outputs (2026): descriptive_statistics.csv, pearson_correlation_matrix.csv, spearman_correlation_matrix.csv, high_correlation_pairs.csv, vif_results.csv, pca_explained_variance.csv, pca_loading_matrix.csv, and accompanying plots (unpublished project analysis).

Appendix — Suggested figure files for team integration

Additional images supplied: correlation_heatmap_full.png; correlation_heatmap_strong.png; numerical_histograms.png; numerical_boxplots.png; variance_comparison.png; std_comparison.png;

pca_scree_plot.png; pca_pc1_vs_pc2.png; strongest_correlations_scatter.png; churn_distribution.png. The report embeds the VIF, PCA cumulative variance and loading figures; any other figure can be inserted by the group as needed. Correct the duplicate-looking class colors in the PCA scatterplot legend if using it.